
Software Design Specifications

for

<Digital Complaint and Grievance System>

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Document Information

Title: Digital Complaint and Grievance System	Document Version No: 1
Project Manager: KASA SAI	Document Version Date:
Prepared By: Group 20	Preparation Date:

Version History

Ver. No	Ver. Date	Revised By	Description	Filename
0.1	2 Mar 2026	Group 20	Created initial wireframes and basic design	
0.2	9 Mar 2026	Group 20	Developed login page, user dashboard, and submit complaint page	
0.3	23 Mar 2026	Group 20	Added auto-detect category feature and admin dashboard	
1.0	6 Apr 2026	Group 20	Connected frontend with backend and database, finalized working system	

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1 Introduction

The Software Design Specification (SDS) provides an overview of the entire software design of the Smart Governance Intelligence System (SGIS). It describes the system architecture, components, and interactions between modules. This document explains how the requirements are transformed into design and serves as a reference for developers and testers

1.1 Purpose

This document describes the detailed design of the **Smart Governance Intelligence System (SGIS)**.

It explains how the system is structured, how components interact, and how the requirements are implemented.

It is mainly useful for:

- Developers (implementation)
- Testers (validation)
- Instructor (evaluation)

1.2 Scope

This design document covers:

- System architecture
- Module breakdown
- Data handling
- Interfaces and workflows

The system is a **web-based complaint management platform**.

1.3 Definitions, Acronyms, and Abbreviations

- SGIS → Smart Governance Intelligence System
- UI → User Interface
- Admin → Administrator
- Officer → Government staff handling complaints
- DB → Database

1.4 References

IEEE SRS & SDS standards
Software Engineering course material
UML Modeling guidelines
Your project SRS document
DBMS concepts (for database design)

2 Use Case View

This section describes the main use cases of the Smart Governance Intelligence System (SGIS). These use cases represent the key functionalities of the system and show how different users interact with it.

The system mainly involves three actors: User, Admin, and Officer. Each actor performs specific actions such as submitting complaints, assigning tasks, and updating complaint status.

2.1 Use Case

UC1 – Submit Complaint

User logs in → fills complaint form → submits → system stores in DB

UC2 – Track Complaint

User enters complaint ID → system fetches status → displays result

UC3 – Assign Complaint

Admin views complaints → assigns to officer

UC4 – Update Complaint Status

Officer logs in → updates complaint → system saves

UC5 – Provide Feedback

User gives feedback after resolution

3 Design Overview

This section provides an overview of the overall software design of the Smart Governance Intelligence System (SGIS). It describes the system architecture, main components, and how different modules interact with each other.

The system follows a layered architecture consisting of the presentation layer, business logic layer, and data layer. The presentation layer handles user interaction through the web interface, the business logic layer processes complaints and manages workflows, and the data layer stores and retrieves information from the database

3.1 Design Goals

- Simple and user-friendly interface
- Fast response time
- Secure access control
- Scalable architecture

Design Constraints

- Requires internet
- Uses relational database
- Role-based access system
- Web-based application

3.2 Design Assumptions

- Users have internet access
- Officers update regularly
- Server availability is high

3.3 Significant Design Packages

1. User Module

- Registration & login
- Submit complaint
- Track complaint

2. Admin Module

- View complaints
- Assign complaints

3. Officer Module

- View assigned tasks
- Update status

4. Database Module

- Stores all system data

3.4 Dependent External Interfaces

The table below lists the public interfaces this design requires from other modules or applications.

External Application and Interface Name	Module Using the Interface	Functionality/Description
Web Browser	HTTP	UI interaction
Database	SQL	Data storage

3.5 Implemented Application External Interfaces (and SOA web services)

The table below lists the implementation of public interfaces this design makes available for other applications.

Interface	Module	Description
Login API	User Module	Authentication
Complaint API	System	Submit & fetch complaints
Status API	Officer Module	Update complaint

4 Logical View

This section describes the internal structure of the Smart Governance Intelligence System (SGIS) and how different components interact to perform system operations.

The system is organized into multiple layers and modules. The main modules include the User module, Admin module, Officer module, and Database module. Each module is responsible for specific functionalities such as complaint submission, assignment, status updates, and data storage.

4.1 Design Model

System follows **3-tier architecture**:

Presentation Layer

- Web pages (UI)

Business Logic Layer

- Complaint handling
- Assignment logic

Data Layer

- Database operations

Main Classes (conceptual)

- User
- Complaint
- Admin
- Officer
- Feedback

4.2 Use Case Realization

Example: Submit Complaint

Flow:

1. User inputs data (UI)
 2. Controller validates data
 3. Complaint object created
 4. Stored in database
 5. Complaint ID generated
- Example: Assign Complaint
6. Admin → selects complaint → assigns officer → DB updated

5 Data View

This section describes how data is stored and managed in the Smart Governance Intelligence System (SGIS).

The system uses a database to store all important information such as user details, complaints, complaint status, and feedback. The main entities in the system include User, Complaint, Officer, and Feedback.

Each complaint is assigned a unique complaint ID and is linked to a user and an officer. The database maintains the status of each complaint from submission to resolution. This structured data model ensures efficient storage, retrieval, and management of information.

5.1 Domain Model

Entities:

- User
- Complaint
- Officer

- Feedback

Relationships:

- User → creates complaint
- Admin → assigns complaint
- Officer → updates complain

5.2 Data Model (persistent data view)

Tables:

- Users (User_ID, Name, Email)
- Complaints (Complaint_ID, Description, Status)
- Officers (Officer_ID, Name)
- Feedback (Feedback_ID, Comment)

5.2.1 Data Dictionary

Field	Description
User_ID	Unique user ID
Complaint_ID	Unique complaint ID
Status	Pending / Resolved
Category	Complaint type
Feedback	User response

6 Exception Handling

- Invalid login → error message
- Empty form → validation alert
- Database failure → retry mechanism
- Unauthorized access → blocked

7 Configurable Parameters

This section describes the parameters that can be adjusted in the Smart Governance Intelligence System (SGIS).

This table describes the simple configurable parameters (name / value pairs).

Parameter	Usage	Dynamic
Session Timeout	Auto logout	Yes
Max Complaints	System limit	Yes
Retry Limit	Error handling	No

8 Quality of Service

This section describes the performance and reliability aspects of the Smart Governance Intelligence System (SGIS).

8.1 Availability

- System available 24/7
- Minimal downtime

8.2 Security and Authorization

- Login authentication
- Role-based authorization
- Secure data storage

8.3 Load and Performance Implications

- Fast page loading
- Handles multiple users
- Efficient DB queries

8.4 Monitoring and Control

- Logs maintained
- Admin monitoring tools
- Error tracking system